Chapter 8
Frontiers of AI in Education

At present, AI in education is one of the most active areas in AI research. It has attracted the attention of researchers in China and abroad. In recent years, many countries have set up research institutions to foster research about AI-powered education. In China, the Advanced Innovation Center for Future Education at Beijing Normal University, the National Digital Learning Engineering Technology Research Center in Central China Normal University, as well as the Collaborative Innovation Center for Educational Informationization, are engaged in academic research and engineering development of intelligent educational services to achieve the strategic objectives of *China's Medium-and Long-term Educational Reform and Development Planning Outline*. In the USA, organizations like National Center for Research on Evaluation, Standards, and Student Testing (CRESST) at University of California, Los Angles, the Learning in Informal and Formal Environments (LIFE) NSF science of learning center hosted at the University Washington in partnership with Stanford University and SRI International, and WestEd are also engaged in research of learning analytics to understand how to use AI technologies to improve the quality of education.

In terms of the publication of research in international conferences like *International Joint Conferences on Artificial Intelligence* (IJCAI), *Conference on Information and Knowledge Management* (CIKM) and academic journals including *Computers & Education, International Journal Journal of Computer-Supported Collaborative Learning*, and *British Journal of Educational Technology*, there are a few articles about how to power education with AI technologies. Besides, Artificial Intelligence in Education (AIED), as a representative international conference about how to integrate AI technologies in education, has attracted researchers and scholars from all over the world.

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This chapter provides a review of current work. It also introduces the possible trends of research in this field, focusing on the following eight major hotspots: knowledge graph, cognitive diagnosis, student modeling, educational questions representation, machine reading comprehension and scoring, intelligent tutoring system, neuroscience and education, and human–machine integration.

8.1 Educational Knowledge Graph

Knowledge representation usually refers to the description of knowledge in the real world into certain forms that the computer can store and utilize. Since the early stage of AI research, there have been studies on knowledge representation. The knowledge graph is an effective way of knowledge representation, which is a multi-relational graph structure of a knowledge base that involves a variety of entities and the associations among them in a particular domain (see Fig. 8.1). A knowledge graph describes basic concepts, entities, and their relations in the objective world in a structured form that closes to the form in human cognition. The knowledge graph provides a better way to organize, manage, and understand massive information [1]. It can support a wide range of applications in AI, including intelligent semantic retrieval, intelligent assistant, and intelligent question and answer service.

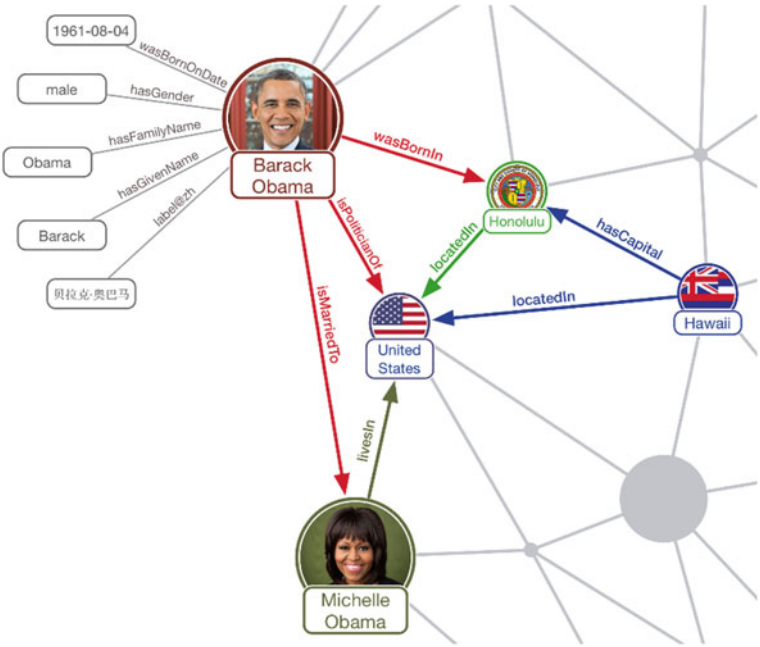

Fig. 8.1 Example of a general knowledge graph

The knowledge graph is usually divided into two categories, general knowledge graph and vertical domain knowledge graph. A general knowledge graph usually contains a large number of real-world commonsense knowledge with extensive coverage. Typical general knowledge graphs include DBPedia, Freebase, and WordNet. The vertical domain knowledge graph, also known as an industry knowledge graph, is usually domain-oriented. The vertical domain knowledge graph can be viewed as an industry knowledge base based on semantic technology. A vertical knowledge graph is typically built with domain data and specific purposes, which is more demanding about the depth and accuracy of knowledge and information constructed.

An educational knowledge graph is a knowledge graph constructed for educational needs. It aims to show the different elements involved in the teaching process and the cognitive relations with educational significance among these elements. In general, the dots in the educational knowledge graph represent different elements in education, in terms of basic concepts, educational objectives, test questions, and subject textbooks. The edges in the graph stand for the relationships between different elements, such as the precursor and successor of basic concepts and the correspondence between test questions and learning objectives. The educational knowledge graph is the core of an intelligent educational system. With an educational knowledge graph, we can organize massive heterogeneous learning resources in a more structured way, construct relationships among basic concepts, support group collaborative learning, provide questions and answers about subject knowledge, and foster adaptive learning and recommendations.

Existing work on knowledge graph in education can be summarized into two categories in terms of different goals: knowledge and curriculum-based graph and teaching resources-centered graph. Research on curriculum-based graphs focuses on the analysis of the relationship between basic concepts and courses. For analysis of basic concepts, some researchers seek to refer to the precursor and successor relationships between basic concepts by analyzing the frequency of different concepts mentioned in textbooks [2]. Others attempt to understand the relationship by analyzing educational resources and students' behavior data [3], as well as the interdependence of different courses [4]. Meanwhile, there are also researchers constructing curriculum concept maps [5, 6] and analyzing the interdependence of different online courses [7] based on the precursor and successor relationship between concepts. However, resources-centered knowledge graphs place a particular emphasis on the elements involved in teaching. For instance, Wei has constructed a knowledge base of terms in the field of educational technology [8]. Zheng and Ying analyze the relationship between different disciplines by constructing a knowledge graph for them [9]. There are also researchers constructing knowledge graph for different disciplines in higher education to analyze teachers' expertise in the college [10]. Besides, organizing learning content knowledge, creating a learning environment, building a semantic social network, integrating and sharing learning resources, and building a learning knowledge base are also a better way to provide learners with appropriate learning services [11].

Although previous research on educational knowledge graphs has made considerable progress [12, 13], existing research on the knowledge graph still lays emphasis on the analysis of a single entity type, such as basic knowledge concepts, curriculum, resources. The field is still lacking a more systematic comprehensive analysis of various factors involved in the teaching process. At the same time, an educational knowledge graph focusing on a single entity type may result in fewer sources of data involved in the analysis. How to address this issue is an urgent challenge in constructing and applying a knowledge graph in education.

8.2 Cognitive Diagnosis

In traditional teaching, teachers' energy and attention are too limited to know about the status of each student at the same time, especially in classrooms with a large number of students. The typical way to evaluate students' learning achievement is to record their sources on test questions, which is improper to know each student's understanding level of each knowledge concept. With the application and popularization of all kinds of information technology and products, a large amount of students' learning data is recorded. The cognitive diagnosis model integrates student's answers to test questions and the basic knowledge concepts that each test question has converted to evaluate students' learning status and mastery levels. This process is called cognitive diagnosis analysis, which plays an essential role in improving the teacher's teaching efficiency.

Cognitive diagnostic analysis, originated in the 1950s, aims to help teachers better understand students' learning status and customize personalized learning strategy for them [14]. Conventional cognitive diagnostic models include item response theory (irt) [15] and deterministic input, noisy And gate (DINA) [16]. IRT is a single-dimensional continuous model. IRT models student's knowledge state into a one-dimensional continuous skill values. Then, it carries out diagnostic evaluation by integrating the factors of test questions (difficulty level, differentiation level, and guessing level). In practice, the IRT model cannot diagnose the value of students' mastery of different skills. Therefore, scholars put forward multidimensional discrete models, the typical representative of which is the DINA model. The DINA model uses the Q matrix to represent the relationship between exercises and basic knowledge concepts. Meanwhile, factors such as "error" and "guess" are taken into account to model students' skills to predict students' mastery or failure of specific knowledge concepts.

In traditional teaching, due to the limitation of data recording means, students' learning data can merely be recorded in the classroom or on campus. The amount of students' learning data is small. Meanwhile, the correlation of the data is relatively weak. However, with the application of information technology and big data technology, the quantity and quality of data collected are continually improving. Data collected shows the features of large sample size and non-isolated. The traditional IRT and DINA models are no longer suitable for this situation. Therefore,

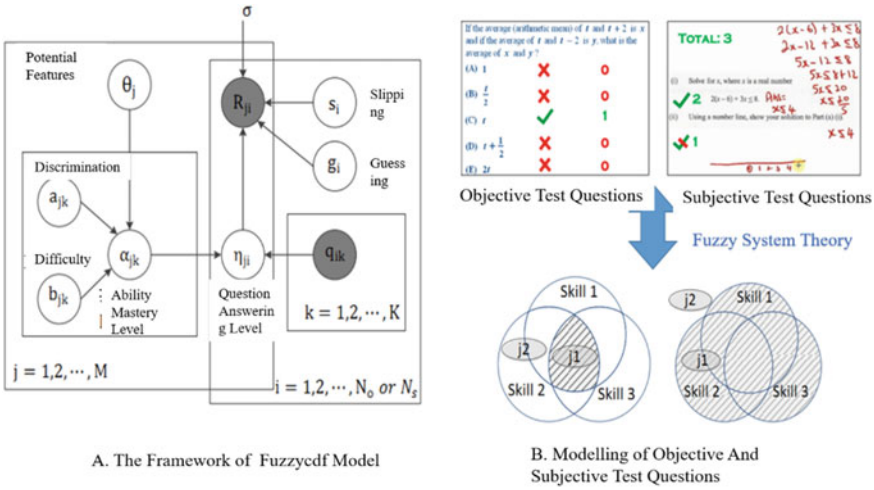

Fig. 8.2 FuzzyCDF model

many big data analysis techniques are also used for cognitive diagnosis. The most common technique is matrix decomposition [17] method, which projects students and exercises into the low-dimensional knowledge of the hidden space and correlates learning data of different students to make sure the diagnosis results among students are comparable. Based on the IRT model, the DINA model, and the matrix decomposition technique, some scholars have put forward other refined models. For example, with regard to the shortcomings of traditional cognitive diagnostic models like small sample size, single skill response mode as well as discrete diagnosis, the fuzzy cognitive diagnosis framework (FuzzyCDF) is proposed to enable collaborative modeling of large sample size and sparse answers [18]. As shown in Fig. 8.2, the framework integrates IRT with fuzzy theory and takes into account factors like test difficulty level, differentiation level, carelessness, guessing, and others. The FuzzyCDF model adopts the skill continuity theory for objective questions (to master all the skills required for exercises) and skills compensation theory for subjective questions (to master one of the skills required for exercise that can be scored). The framework sets probabilistic parameters the mastery levels (not just mastery or not-mastered) as well. All these adjustments make the diagnosis results with increased granularity and accuracy. To address low-dimensional implicit space interpretation in matrix decomposition, the exercise and basic knowledge concepts correlation matrix are integrated into the model to refine the explanatory level of diagnosis result and make personalized recommendations [19].

8.3 Student Modeling

In recent years, the concept of learning analytics has been proposed in educational technology, which aims to analyze the learning process with data-driven research based on objective data of learners and learning environments [20]. The ultimate goal is to optimize the learning process and learning environments. Student modeling is an essential direction of research in learning analytics. In general, student modeling is a quantification and abstract representation of students in terms of knowledge, cognitive state, and emotional experience [21]. On the one hand, individual learners can be described quantitatively to meet the needs of personalized and adaptive learning. On the other hand, the student model can portray large-scale learners qualitatively, which provides essential support for teachers and researchers.

The recognition and diagnosis of a learner's knowledge state (student model) are one of the most critical directions in traditional student modeling. This student model focuses on the tracking of the knowledge state changes of learners in learning. Previous research on student modeling focuses on the Bayesian knowledge tracing (BKT) model [22] and the deep knowledge tracing (DKT) model [23]. The BKT model primarily uses the hidden Markov model to solve the tracking problems in education. It introduces the probability definition of slipping and guessing in a targeted manner. In this model, BKT assumes that the students' mastery of knowledge is a binary variable. "0" indicates no mastery; "1" stands for mastery. Using the learner's evaluation data as the sequence training data, the parameters in this model can be learned to obtain students' knowledge state [24]. On this basis, subsequent research expands the BKT model by considering students' prior knowledge [25] as well as the difficulty level of the topic [26, 27]. However, BKT's binary hypotheses are lack of quantitative representation of knowledge mastery level. The feature of the hidden Markov model also results in the limitation of the BKT—only a single knowledge concept can be evaluated. Multiple knowledge concepts cannot be analyzed at the same time. Therefore, some researchers put forward a DKT model based on deep learning. Based on the RNN structure, the DKT model uses neural networks to keep track of the time series information in the learner's assessment data to predict the changes of students' knowledge state. Aiming to address the limitation of the BKT model, the implicit variable of the DKT model represents learners' mastery of different knowledge concepts. It uses the continuous value between 0 and 1 to represent learners' proficiency in each knowledge concept. Based on learners' time series evaluation data, the DKT model can identify learners' knowledge state more comprehensively and accurately. On this basis, subsequent research adds the feature of distributed learning of exercises [28] as well as the in-depth knowledge tracking model that considers the precursor relationship between different knowledge concepts [29] (as shown in Fig. 8.3).

On the other hand, previous research also finds out that learning motivation, concentration, and emotional change usually directly affect the learning effect and quality in the learning process. Constructivism theory, for example, holds that learners' emotional states may systematically influence the way they process new

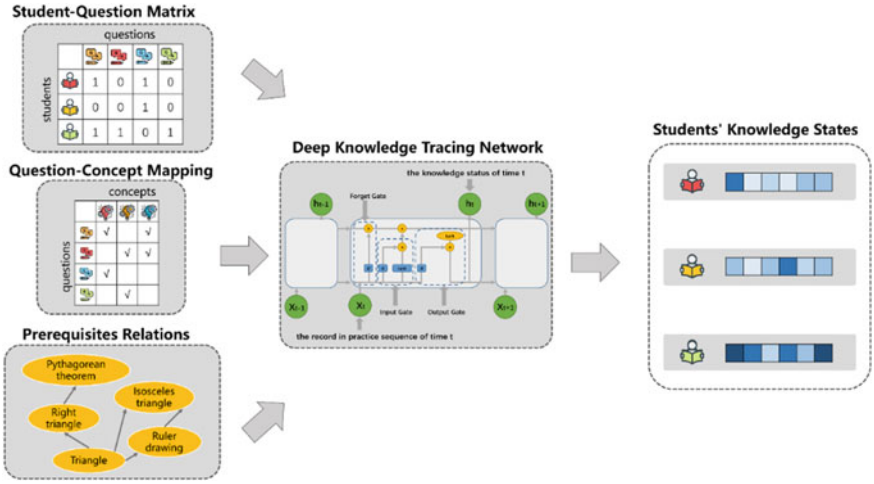

Fig. 8.3 Deep knowledge tracing framework

data [30]. Meanwhile, there is also research indicating that different levels of learning focus can also be inferred effectively through specific behaviors (overt behaviors) in the learning process [31]. Researchers used facial recognition technology to investigate the emotional state of learners in the learning process [32]. Besides, a variety of devices, such as intelligent wearable devices, a pressure-sensitive mouse, and skin electrical conduction sensors, are also used in student modeling. For example, smart watches are used to monitor the behaviors of students in the classroom (raising hands, taking notes, etc.) to effectively predict and intervene in students' learning [33]. At the same time, researchers used the interaction data of large-scale learners in the learning process and the learners' learning achievement data at different stages to model learners through the classification and regression algorithms in machine learning, such as the dropout prediction model [34] based on the MOOC platform and learners grouping model [35].

Student modeling is one of the most critical directions for applying AI technologies in education. Currently, how to accurately describe and quantify learners' complicated knowledge status and knowledge structure is still one of the hotspots and puzzles of intelligent education research. With the development of AI technology, the data foundation and construction mode of the learner model is changing gradually. The development of large-scale online learning environments and intelligent sensor technology can help to effectively collect a large amount of learning behaviors and learning process data. At the same time, technologies such as natural language processing, emotional computing, and deep learning can also help extract and analyze the data collected automatically. Finally, with an accurate student model, the intelligent tutoring system can offer learners more appropriate service and personalized help.

8.4 Educational Question Representation

The representation of educational question resources is the premise and foundation of machine reading comprehension, the construction of educational test question association network, student's score prediction, and personalized resource recommendation. The first issue of the deep representation, connection, and application of large-scale educational resources is about how to make a reasonable and effective semantic representation of heterogeneous multi-modal educational questions. In recent years, online learning systems in China have accumulated a large number of educational issues with features like multi-source and heterogeneous (see Fig. 8.4), sparse and high noise, as well as with rich semantic information. For example, educational questions are typically composed of text, images, and basic knowledge concepts. How to integrate multi-source heterogeneous data using multi-modal learning methods and how to understand the purpose of educational questions and the corresponding semantic representation of them are pressing challenges to be addressed in research. Traditional methods do not take any information in educational questions into account. Those methods merely carry on statistical analysis to the label information inherent in educational questions (such as basic knowledge concepts, the difficulty level, and the differentiation level) and take it as the representation of the educational questions.

More importantly, the label information of educational questions is often collected through educational experts manually, which may have disadvantages like inefficiency and inaccuracy.

To this end, relevant research teams seek to use traditional machine learning methods to represent educational questions to further predict and automatically label the questions. Some scholars have proposed to include the reverse propagation of artificial neural networks and the use of five-dimensional eigenvectors (including the relevance of test text, average word frequency, test length, answer length, and the variance of word frequency distribution), while the disadvantage is the lack of semantic analysis of educational questions [36]. Researchers further analyze and compare the performance of the nearest neighbor algorithm, the naive Bayesian, as well as the support vector machine in the representation of educational questions [37] and enhance the performance of representation by extracting the entries, semantics,

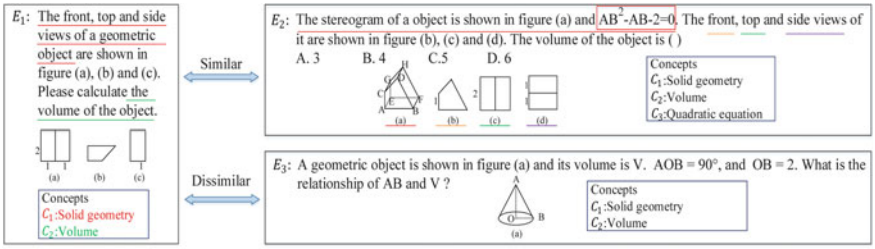

Fig. 8.4 Multi-source heterogeneous education data

keywords and syntax information from educational questions [38]. On this basis, researchers further suggest the use of the extreme learning machine model instead of the support vector machine model to better represent educational questions [39].

The traditional method of question difficulty prediction relies on educators' and researchers' expertise to evaluate the difficulty level of educational questions manually or use simple regression models in educational psychology. The former method suffers from the expert's subjectivity. Moreover, it is time-consuming. One disadvantage of the latter approach is that it involves a considerable number of samples and the work of representation of text features of educational test questions. Therefore, researchers put forward new solutions, such as the TACNN framework [40] and other deep learning models to represent multi-source heterogeneous education resources. The TACNN model uses the text information of questions and the record of the students' previous answers to predict the difficulty level of the reading comprehension test in the future. In this model, a unified way was designed to represent the semantic information in each part of the question and quantify the feature representations. On the other hand, in the college entrance examination in China and other standardized examinations like SAT and GRE, it is unavoidable that some test questions could be guessed by test-takers. To address this issue, some researchers put forward the MANN model, which is applicable to large-scale educational resources [41].

As shown in Fig. 8.5, the model makes full use of the multi-modal feature (i.e., text, images, basic knowledge concepts) of the educational questions and masters the uniform semantic representation of each educational question. It also puts forward an attention model to capture the association between text-concept and text-image in each question. Besides, a similarity attention model is designed to measure the similarity between each question and its semantic representation.

In general, with the most cutting-edge technologies of AI, including word embedding, CNN, RNN, and attention, we can label and represent most educational test questions automatically and continuously improve the results in use as feedback to

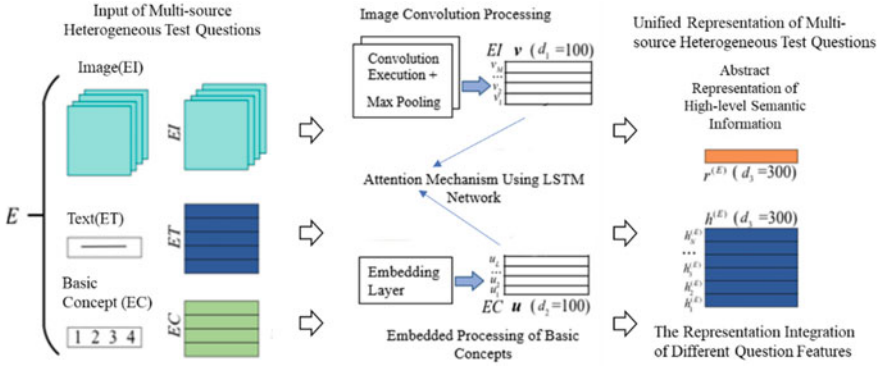

Fig. 8.5 MANN model

the model. By self-correcting and learning, the correct rate of labeling will be further improved to the level of advanced teachers.

8.5 Machine Reading Comprehension and Scoring

It is another essential direction of AI research for machines to read and respond to educational questions in different disciplines. Overall, the goal is to train machines to complete automatically reason and extract the appropriate answers based on the given questions and relevant documents.

Research on machine reading comprehension falls into the following categories, including algebraic [42] and geometry questions [43], science questions [44], and Chinese reading comprehension [45]. Traditional research on reading comprehension and automatic answering mainly relies on the method of feature engineering and combines the non-hidden variables [46] or the hidden variable models [47, 48].

This approach searches for answers to specific questions through the calculation of eigenvectors and the semantic similarity of the documents.

In recent years, research on machine reading comprehension based on deep learning has also been put forward. It is promising based on the results in the testing of the large-scale corpus. It is unnecessary to conduct sophisticated feature engineering manually for the deep learning model. The model completes the feature extraction automatically in the training process. At present, the two conventional deep learning models are the convolution neural network (CNN) and the recurrent neural network (RNN). Neurons that the RNN model commonly use is the long short-term memory (LSTM) unit, which solves the problem of gradient disappearance in RNN training well. The training algorithm of the RNN network is based on error reverse propagation algorithms, such as the back propagation through time algorithm (BPTT) [49]. Other researchers further introduced the attention mechanism to improve the deep neural networks in reading comprehension [50, 51] as shown in Fig. 8.6. The attention mechanism can better learn the location distribution of relevant information about a specific question in a document. It also can help to shape a more adequate representation of semantic features [52, 53].

Currently, there are many large datasets established to evaluate the models developed for reading comprehension and automatic answering. Among them, SQuAD [54] is a multi-domain dataset based on Wikipedia (see Fig. 8.7). One concern about this dataset is that there is no given candidate answer. The answer is typically a continuous fragment in the original text, including numbers, entities, phrases, and sentences. Researchers in China and abroad have designed many models of machine reading comprehension for SQuAD. Some of these models show excellent performance in reading comprehension, such as the R-NET model from the natural language computing group of Microsoft Research Asia on January 3, 2018. The exact match (EM) value reaches 82.650, which goes beyond the score rated by a human (82.304). Datasets that are commonly used include TtriviaQA [55] and MCTest [56].

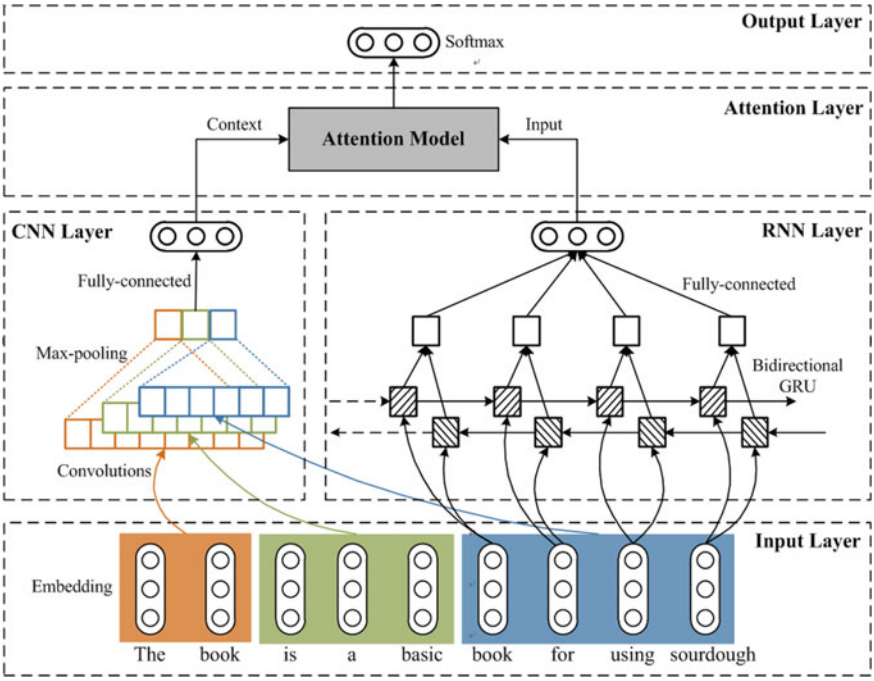

Fig. 8.6 Deep neural network model

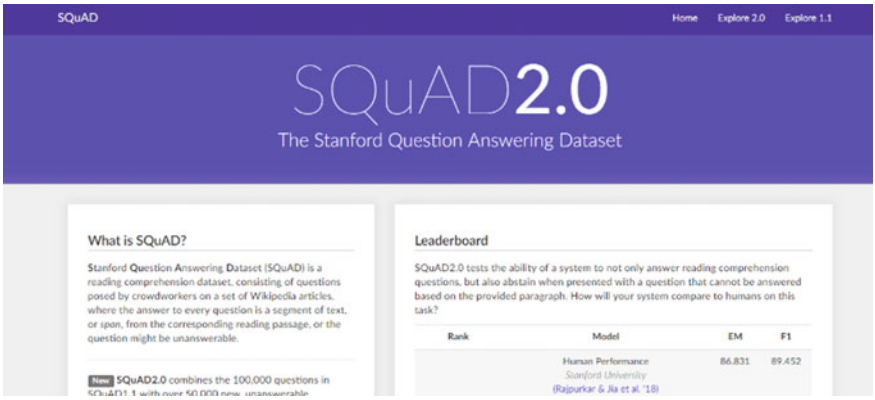

Fig. 8.7 SQuAD dataset of reading comprehension

In addition, in order to achieve machine reading comprehension and automated scoring, we need to de-noise some relevant documents [57], process the answers from different areas with more detailed information, and evaluate the answers with different methods [58, 59]. The vector representation of the answer can be further analyzed. For the answers that are above a certain threshold, the machine will give it

a score on its own. For the ones that are below a certain threshold, it will be scored manually.

8.6 Intelligent Tutoring System

In the 1980s, with the second upsurge of AI, a large number of studies on the intersection of cognitive science and computer science emerged in the field of computer science and cognitive science [60, 61]. Researchers build intelligent systems to understand and validate the results of experiments in cognitive science. As the second wave of AI recedes, there is less cooperation between AI scientists and cognitive scientists. AI scientists and cognitive scientists formulated their fields separately. However, the initial cooperation between them still laid a foundation for research on intelligent education. The intelligent tutoring system produced as a product of collaboration integrates not only computer technology but also the principles of students' cognitive development, which can play an essential role in solving practical problems in education. The fundamental components of the traditional intelligent tutoring system are shown in Fig. 8.8.

Since the 1990s, many successful intelligent teaching systems have been popularized and demonstrated for their effectiveness in teaching different subjects. For example, there are Andes who teaches students physics problem solving [62], Algebra tutor which teaches students algebra [63], AutorTutor which teaches students information technology, physics, biology, and other subjects through dialogue [64], as well as Crystal Island which assists students in exploring and learning in a virtual 3D environment [65]. In terms of the different roles that the tutoring systems play in supporting students' learning, the intelligent tutoring system evolves from supporting practice and problem solving to include a variety of different ways of learning, which are not limited to:

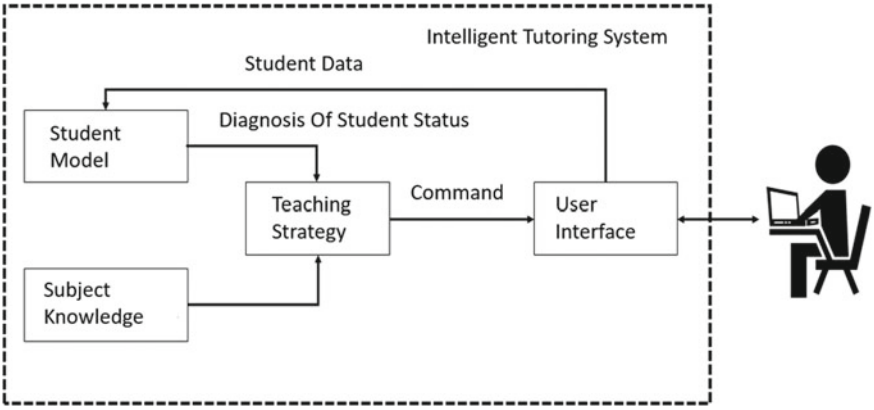

Fig. 8.8 Components of the traditional intelligent tutoring system

- Design intelligent agents that could be taught by students to support their learning [66]. With the development of robot technology, some researchers create environments that students teach physical robots to consolidate their mastery of knowledge [67, 68].
- Many researchers have proposed the theory of game-based learning. Through the design of educational games, students can learn in a more relaxed and happy manner [69].
- It allows students to learn in the form of dialogue in natural language. Valuable knowledge is incorporated into the dialogue to enable learning through dialogue [70].
- It helps students to deeply understand the dynamic relationships between different variables in an authentic context by creating a dynamic model [71].

Theories of education and cognitive sciences are integrated into each intelligent tutoring system to guide agents in selecting appropriate questions, providing timely feedback, and tracking the changes in students' status by establishing corresponding student models. In general, finite state machine could be used to encode student behaviors and to serve as a detector for student-specific strategies. For example, the Bayesian statistical models can be used to keep track of changes in students' status of knowledge mastery level.

With the development of AI technologies, the input of an intelligent tutoring system gradually shows the feature of multi-modality. The initial intelligent tutoring system can merely identify the pre-defined text content. With the improvement of natural language processing technology, more and more intelligent tutoring systems can understand students' complicate text responses and provide more accurate feedback [72]. With mature speech recognition technology, we can accurately translate students' voices into text, which enables the machine to understand students' voice information and provide appropriate content in the form of dialogue [73]. Interactions between students and the intelligent tutoring system are widely used to evaluate students' knowledge status, problem-solving ability, and metacognitive state [74]. Various popular data mining methods have been used to discover behavioral patterns, to find the connection between behavior patterns and students' learning performance, and to trigger feedback on specific behaviors. With the development of intelligent hardware, various sensors (including skin conductance, emotion recognition, mouse pressure, etc.) have been adopted in the tutoring system. These sensors are used to capture students' emotional states so that the system can offer appropriate teaching guidance according to students' different emotional states [75].

Generally speaking, theories of educational psychology and cognitive science need to be deeply integrated into intelligent tutoring systems. Meanwhile, with the introduction of new technology in the field of AI and more multi-modal interaction data collected from teachers and students, the intelligent tutoring system can make more appropriate decisions and knowledge delivery. At present, the intelligent tutoring system and related products mainly solve some relatively independent problems in educational practice, such as automatic scoring and classroom progress monitoring. These systems and products enable a better understanding of

how students learn and how to foster more effective and deepening learning. Development of the future of intelligent teaching systems needs to further integrate the theories of learning from different fields. Moreover, more efficient and universal intelligent tutoring systems and innovative solutions to solve authentic problems in education are in need.

8.7 Neuroscience and Education

In recent years, the combination of neuroscience and education has been highly concerned in China and abroad. From the perspective of national policies, the USA, the European Union, Japan, and other countries have introduced plans or strategies for the development of neuroscience and invested a large number of funds in supporting the research and development of neuroscience. China has also included the "China Brain Plan" as a significant project in the 13th Five-Year Plan. The overall framework of the "China Brain Program" (see Fig. 8.9) will rely on a deep understanding of the neural basis of the cognitive function of the brain.

This program seeks to promote the development of neuroscience and improve public health through the research and development of brain-computer technology as well as the methods for diagnosis and treatment of brain diseases [76]. In terms of research on neuroscience, researchers are carrying out interdisciplinary research that combines findings from neuroscience and education. Some researchers have suggested that the integration of young international researchers who dedicated to neuroscience and education is a valid path to the development of educational neuroscience [77]. At present, many international research centers begin to pay close attention to interdisciplinary research, such as research on reading and writing skills, computing ability, emotion, and learning. There are also studies that begin to use neuroscience to explore the predictive effects of neural signals on students' learning performance [78] as well as the mechanisms of teacher–student interactions [79, 80].

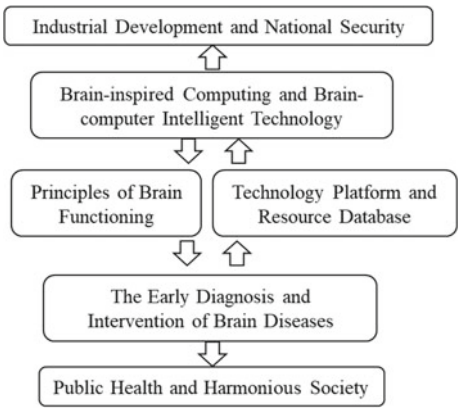

Fig. 8.9 Framework of the brain plan in China

Concerning its practical application in educational practice, a large number of classroom studies seek to explore the impact of neuroscience-based teaching methods [81, 82].

Many scholars believe that it is too early for neuroscience science to have an impact on education [83]. Some of them believe that it is impossible to promote classroom teaching and learning directly [84]. A growing number of researchers believe that neuroscience-related research can be more productive or have a clear advantage over traditional education to predict what matters for better individual learning [85]. In a meta-analysis study that reviews 31 educational studies on neuroscience-based learning from different countries from 1999 to 2011, researchers found that 83.34% of the studies show positive effects, which confirms that the teaching and learning based on neuroscience is conducive to the improvement of students' academic achievement [86].

Except for the focus on interdisciplinary research, a small number of researchers began to examine teachers' perception of the "nerve myths" [87], the conception of neuroscience [88], and teachers' neuroscience literacy [89]. Teachers' knowledge, belief, and practice of neuroscience are most favorable to promote the adoption of neuroscience in education. At present, some colleges and research institutions have started to set up programs and courses in educational neuroscience and initiated projects to promote their development. However, there are still very few researchers concerned about teachers' literacy of neuroscience in China and abroad.

To sum up, there are still many challenges and issues in integrating the findings of neuroscience into education. There is research indicating that research of neuroscience and education can help each other to move forward. On the one hand, findings from neuroscience research can provide direct evidence in classrooms. On the other hand, research findings applied to education can gain more practical value [90]. At present, the field of educational neuroscience has attracted the attention of policy decision-makers, researchers, and educators. The international collaboration, interdisciplinary integration, and co-development of theoretical and empirical research are the major characteristic. Educational neuroscience will bring about rich perspectives on educational practice. It will also serve as the foundation of AI technologies.

8.8 Human–Machine Integration

At present, AI technology based on machine computing has made breakthroughs in many areas. However, it is still challenging to construct strong AI with standard attributes to enable core intelligence like automated decision-making reasoning about causality. Therefore, how to integrate humans and machines to enable deep collaboration is an important direction. On the one hand, researchers are expecting the integration of brain and machine with the help of neural technology [91, 92]. On the other hand, human-based computation (HBC) is proposed as a pathway to explore the integration of machines with human beings [93].

There is a big difference between human and machine intelligence. People can use their perceptual cognition to make rational sampling on the objective physical world and summarize the rules and logic based on their perceptual cognition. However, current machine intelligence is still fundamentally based on a series of mathematical logic and symbols of an utterly rational representation. Therefore, due to the complex cognitive process that cannot be fully represented mathematically, the cognition of AI is not real to some degree. In short, machines have advantages over human beings in terms of search, computation, and storage. However, humans outperform machines in terms of perception, reasoning, and learning. Therefore, it is necessary to combine the intelligence and advantages of both the machine and human beings to design a new intelligent system that integrates human beings, machines, and environments.

Research of human–machine integration examines the mixture of physical and biological systems. Firstly of all, there is a need to establish a two-way interaction between people and machines. According to the overall information theory and other relevant theories [94], an intelligent system must be highly integrated. Secondly, it is necessary to integrate machine intelligence using group behavior characteristics, structural characteristics, and interactive features as well as decision-making [95]. Ultimately, there is a need to investigate how to enable human–machine intelligence collaboration through an appropriate paradigm [96]. For example, how to make people consciously participate in and integrate their identification, association, reasoning and other abilities into a particular computing task, or how to use the rules of behavior that people behave unconsciously as intelligence to accomplish specific computational tasks. At the same time, how to divide and assign the computational tasks correctly and determine the right order of performing sub-tasks are also pressing issues in this field.

Generally speaking, research on human–machine integration is still in its infancy. It is the product of the development of AI technologies to a certain extent. With the further research of human cognition mechanisms in neuroscience and education, we can introduce the human–machine model to AI to coordinate the development of the two to improve the intelligence of machines in reasoning and decision-making, which will become one of the shared objectives of education and AI.

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